

Continual, Format-Agnostic Ability Tracking

Dynamic IRT, tested on Duolingo SLAM

Wenrui Yuan

June 2026

The assessment we have, and the one learners need

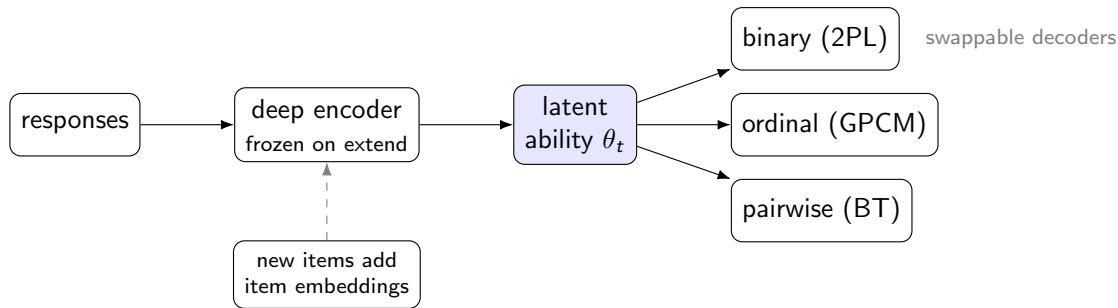
Most learner assessment today is

- **Binary.** Right or wrong, discarding the partial-credit signal in *how* a learner answers.
- **Static.** Ability treated as fixed, estimated in batch over a full response matrix.
- **Per domain.** A separate model for each subject, with little shared structure.
- **Recalibration-heavy.** New content forces re-estimating the whole scale.

Yet learners answer in grades, ability moves as they learn, and platforms now span language, math, music, and chess.

A single learner-state model

A deep sequence encoder turns a response history into a latent ability. Swappable IRT decoders read that ability into any response format. New items attach to the existing scale. Ability is tracked as it moves.



IRT-flavored (interpretable ability, discrimination, step thresholds) and tracking-flavored (dynamic, sequence-based), constrained by neither.

Three readouts, one scale

The encoder maps a response history to a time-varying ability,

$$\theta_t = f_\phi(x_{1:t}), \quad x_t = (\text{item}_t, \text{response}_t).$$

Each decoder is a logistic model of a difference on that one scale, with $\sigma(z) = 1/(1 + e^{-z})$:

$$\text{binary (2PL): } P(\text{correct}) = \sigma(\alpha_j(\theta - \beta_j))$$

$$\text{ordinal (GPCM): } P(r=k) = \frac{\exp \sum_{c=0}^k \alpha_j(\theta - \beta_{j,c})}{\sum_{m=0}^{K-1} \exp \sum_{c=0}^m \alpha_j(\theta - \beta_{j,c})}$$

$$\text{pairwise (BT): } P(i \succ j) = \sigma(s_i - s_j)$$

The pairwise model is Rasch between two items, and Rasch is the binary model between a person and an item. One scale, three readouts.

How we test it, and how we measure

Two evaluation modes, every result labeled as one.

- **Synthetic recovery.** Responses generated from *known* item parameters and a *known*, moving ability. Check the model recovers them. Exact ground truth.
- **Real transfer on Duolingo SLAM.** Whether item difficulty measured one way predicts it measured another, on real learners.

We report **rank recovery** (Spearman) throughout. For item selection and feedback, what matters is whether the model *orders* items and abilities the way the truth does, not whether the numbers match exactly. Rank recovery measures exactly that, and is robust to the skewed difficulty distributions real data shows.

Result 1. Track ability as it moves

synthetic. Each learner's ability follows a random walk, $\theta_{t+1} = \theta_t + \varepsilon_t$, over their answer sequence. The model sees only the responses.

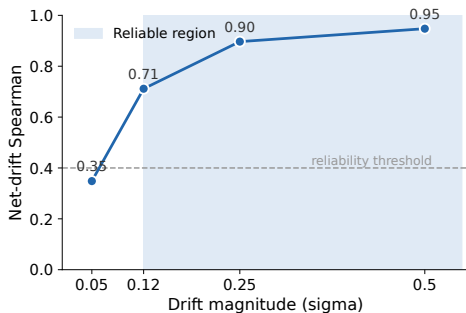


Recovered trajectories follow the true ones. The model infers where ability is heading, not just where it is.

Metric: net-drift rank recovery $\rho_s(\hat{\Delta}, \Delta)$, where $\Delta_p = \theta_{p,T} - \theta_{p,1}$ is each learner's net change. It isolates whether we track the *movement*, separate from the static level.

How much movement the tracking needs

synthetic. Sweeping the size of the drift.

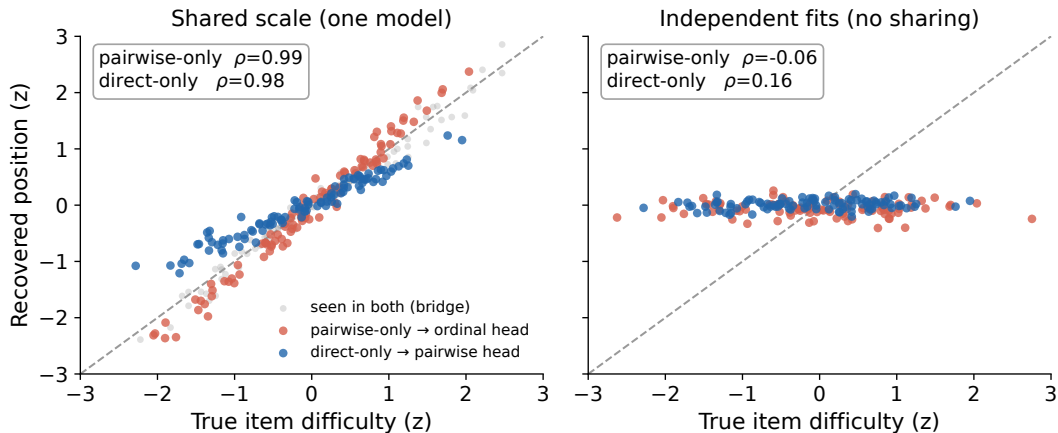


Reliable wherever ability meaningfully moves ($\text{drift} \geq 0.12$). The gain is precisely in the moving regime, where a static estimate cannot reach.

Same net-drift metric as Result 1, swept across how fast ability changes. Low values at near-static are expected, there is barely any change to rank.

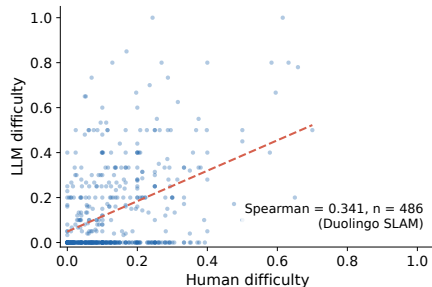
Result 2. One scale across response formats

synthetic. One shared item representation, an ordinal (GPCM) head and a pairwise (Bradley-Terry) head, trained jointly. Some items appear *only* in ordinal data, some *only* in pairwise data, the rest in both.



Result 3. One scale across respondents

Duolingo SLAM, real data. A pool of small language models attempt 486 SLAM tap-format items. We grade them and ask whether their item difficulty ranks human item difficulty.



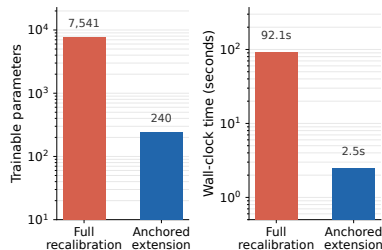
Spearman 0.34. A moderate, real signal from a near-homogeneous model pool, so a lower bound a more diverse pool would likely raise.

Metric: Spearman between model and human item difficulty (proportion incorrect). Tap-format items are used because there the learner selects from given word tiles, so SLAM's exact-match grading artifact is structurally absent.

Result 4. Extend the bank cheaply

synthetic. New items E attach to the frozen scale by fitting only their own embeddings, the classical fixed-anchor calibration of Kolen and Brennan, reused, not reinvented:

$$\min_{\{e_j: j \in E\}} \sum_{j \in E} \sum_p -\log P(r_{pj} \mid \theta_p, e_j), \quad \phi, \{e_j: j \in B\} \text{ frozen.}$$



About 30 times fewer parameters than a full recalibration, and the scale stays stable across repeated rounds with no compounding.

The novelty is not the freezing, which is decades old. It is that the frozen scale is a deep encoder that keeps

What is new, against traditional IRT and CAT

Fixed-parameter calibration, freezing the scale and anchoring new items, is standard in classical adaptive testing (Kolen and Brennan). We use it. We do not claim it.

What is new is where the scale lives, and what it does. In classical CAT the scale is a few explicit item parameters, trivial to freeze, and ability is a static estimate on a fixed item pool. Here the scale is a deep encoder, and the frozen encoder

- tracks a **moving** ability, not a static score,
- **generalizes** to items it never trained on, and
- carries **ordinal and pairwise** responses on one scale, not only right or wrong.

Classical CAT has no encoder to keep tracking, no moving ability, and no shared format. The freezing is borrowed plumbing. The dynamic, generalizing, format-shared scale is the contribution.

The numbers

Result	Metric	n	Data	Value
Dynamic tracking	net-drift rank recovery	1500	synthetic	0.71–0.95
Cross-format transfer	held-out item on other scale	300	synthetic	0.98–0.99
Respondent transfer	model vs human difficulty	486	Duolingo SLAM	0.34
Cheap extension	params vs full recalibration	—	synthetic	$\sim 30\times$ fewer
Continual extension	scale drift over 5 rounds	—	synthetic	none

All correlations are Spearman rank recovery. Synthetic results are a proof of concept; the real-data transfer is moderate and lower-bounded by a near-homogeneous model pool.

Why this fits a multi-domain platform

A platform spanning language, math, music, and chess faces one learner-modeling problem repeated many times.

- One model measures all of them on a single interpretable, extensible scale.
- Ordinal grades and longitudinal tracking, the parts a dichotomous line leaves open.
- Cheap cold-start for the constant stream of new content.

Status and a concrete next step

Now. A single working model, validated as a proof of concept on synthetic data and on a real Duolingo SLAM transfer test, with its reliable operating envelope mapped.

A concrete step together.

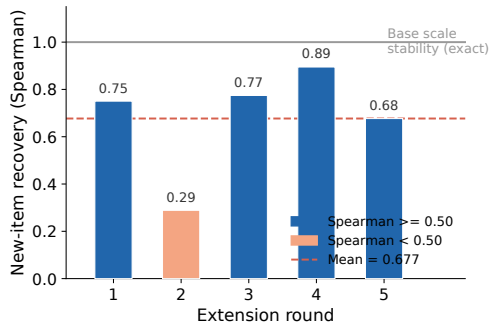
- A slice of partial-credit or ordinal response data, for example graded math or music items, to test the ordinal and longitudinal claims on real learners.
- Or a short pilot co-designing a diverse model pool to sharpen the respondent-transfer result.

References

- Kolen and Brennan. Test Equating, Scaling, and Linking. Springer, 2004.
- Duolingo SLAM shared task. Second Language Acquisition Modeling, 2018.
- AutoIRT. Automated item calibration. Duolingo, 2024.
- BanditCAT. Bandit-based adaptive testing for language assessment. Duolingo, 2024.
- ma-irt. Deep ordinal item response theory, our groundwork, under review.

Backup. Extension stays stable across rounds

synthetic. Five sequential rounds, one hundred new items.



The base scale is untouched (stability exact), and per-round new-item recovery stays flat. Error does not compound.